



contents

Chapter 1 Fundamentals I

Development and Properties of C++	2
Object-Oriented Programming	4
Developing a C++ Program	6
A Beginner's C++ Program	8
Structure of Simple C++ Programs	10
Exercises	12
Solutions	14

Chapter 2 Fundamental Types, Constants, and Variables I5

Fundamental Types	16
Constants	22
Escape Sequences	26
Names	28
Variables	30
The Keywords <code>const</code> and <code>volatile</code>	32
Exercises	34
Solutions	36

Chapter 3 Using Functions and Classes 39

Declaring Functions 40
Function Calls 42
Type void for Functions 44
Header Files 46
Standard Header Files 48
Using Standard Classes 50
Exercises 52
Solutions 54

Chapter 4 Input and Output with Streams 57

Streams 58
Formatting and Manipulators 60
Formatted Output of Integers 62
Formatted Output of Floating-Point Numbers 64
Output in Fields 66
Output of Characters, Strings, and Boolean Values 68
Formatted Input 70
Formatted Input of Numbers 72
Unformatted Input/Output 74
Exercises 76
Solutions 78

Chapter 5 Operators for Fundamental Types 81

Binary Arithmetic Operators 82
Unary Arithmetic Operators 84
Assignments 86
Relational Operators 88
Logical Operators 90
Exercises 92
Solutions 94

Chapter 6 Control Flow 95

The while Statement 96
The for Statement 98
The do-while Statement 102
Selections with if-else 104
Else-if Chains 106
Conditional Expressions 108
Selecting with switch 110
Jumps with break, continue, and goto 112
Exercises 114
Solutions 116

Chapter 7	Symbolic Constants and Macros	119
	Macros	120
	Macros with Parameters	122
	Working with the #define Directive	124
	Conditional Inclusion	126
	Standard Macros for Character Manipulation	128
	Redirecting Standard Input and Output	130
	Exercises	132
	Solutions	134
Chapter 8	Converting Arithmetic Types	139
	Implicit Type Conversions	140
	Performing Usual Arithmetic Type Conversions	142
	Implicit Type Conversions in Assignments	144
	More Type Conversions	146
	Exercises	148
	Solutions	150
Chapter 9	The Standard Class string	153
	Defining and Assigning Strings	154
	Concatenating Strings	156
	Comparing Strings	158
	Inserting and Erasing in Strings	160
	Searching and Replacing in Strings	162
	Accessing Characters in Strings	164
	Exercises	166
	Solutions	168
Chapter 10	Functions	171
	Significance of Functions in C++	172
	Defining Functions	174
	Return Value of Functions	176
	Passing Arguments	178
	Inline Functions	180
	Default Arguments	182
	Overloading Functions	184
	Recursive Functions	186
	Exercises	188
	Solutions	191
Chapter 11	Storage Classes and Namespaces	197
	Storage Classes of Objects	198
	The Storage Class extern	200

The Storage Class static	202
The Specifiers auto and register	204
The Storage Classes of Functions	206
Namespaces	208
The Keyword using	210
Exercises	212
Solutions	216

Chapter 12 References and Pointers 221

Defining References	222
References as Parameters	224
References as Return Value	226
Expressions with Reference Type	228
Defining Pointers	230
The Indirection Operator	232
Pointers as Parameters	234
Exercises	236
Solutions	238

Chapter 13 Defining Classes 243

The Class Concept	244
Defining Classes	246
Defining Methods	248
Defining Objects	250
Using Objects	252
Pointers to Objects	254
Structs	256
Unions	258
Exercise	260
Solution	262

Chapter 14 Methods 265

Constructors	266
Constructor Calls	268
Destructors	270
Inline Methods	272
Access Methods	274
const Objects and Methods	276
Standard Methods	278
this Pointer	280
Passing Objects as Arguments	282
Returning Objects	284
Exercises	286
Solutions	290

Chapter 15 Member Objects and Static Members 297

Member Objects 298
Member Initializers 300
Constant Member Objects 302
Static Data Members 304
Accessing Static Data Members 306
Enumeration 308
Exercises 310
Solutions 314

Chapter 16 Arrays 321

Defining Arrays 322
Initializing Arrays 324
Arrays 326
Class Arrays 328
Multidimensional Arrays 330
Member Arrays 332
Exercises 334
Solutions 338

Chapter 17 Arrays and Pointers 349

Arrays and Pointers (1) 350
Arrays and Pointers (2) 352
Pointer Arithmetic 354
Arrays as Arguments 356
Pointer Versions of Functions 358
Read-Only Pointers 360
Returning Pointers 362
Arrays of Pointers 364
Command Line Arguments 366
Exercises 368
Solutions 372

Chapter 18 Fundamentals of File Input and Output 379

Files 380
File Streams 382
Creating File Streams 384
Open Modes 386
Closing Files 388
Reading and Writing Blocks 390
Object Persistence 392
Exercises 394
Solutions 398

Chapter 19 Overloading Operators 411

- Generals 412
- Operator Functions (1) 414
- Operator Functions (2) 416
- Using Overloaded Operators 418
- Global Operator Functions 420
- Friend Functions 422
- Friend Classes 424
- Overloading Subscript Operators 426
- Overloading Shift-Operators for I/O 428
- Exercises 430
- Solutions 432

Chapter 20 Type Conversion for Classes 441

- Conversion Constructors 442
- Conversion Functions 444
- Ambiguities of Type Conversions 446
- Exercise 448
- Solution 450

Chapter 21 Dynamic Memory Allocation 453

- The Operator new 454
- The Operator delete 456
- Dynamic Storage Allocation for Classes 458
- Dynamic Storage Allocation for Arrays 460
- Application: Linked Lists 462
- Representing a Linked List 464
- Exercises 466
- Solutions 468

Chapter 22 Dynamic Members 477

- Members of Varying Length 478
- Classes with a Dynamic Member 480
- Creating and Destroying Objects 482
- Implementing Methods 484
- Copy Constructor 486
- Assignment 488
- Exercises 490
- Solutions 492

Chapter 23 Inheritance 499

- Concept of Inheritance 500
- Derived Classes 502

Members of Derived Classes	504
Member Access	506
Redefining Members	508
Constructing and Destroying Derived Classes	510
Objects of Derived Classes	512
Protected Members	514
Exercises	516
Solutions	520

Chapter 24 Type Conversion in Class Hierarchies 529

Converting to Base Classes	530
Type Conversions and Assignments	532
Converting References and Pointers	534
Explicit Type Conversions	536
Exercises	538
Solutions	540

Chapter 25 Polymorphism 543

Concept of Polymorphism	544
Virtual Methods	546
Destroying Dynamically Allocated Objects	548
Virtual Method Table	550
Dynamic Casts	552
Exercises	554
Solutions	558

Chapter 26 Abstract Classes 565

Pure Virtual Methods	566
Abstract and Concrete Classes	568
Pointers and References to Abstract Classes	570
Virtual Assignment	572
Application: Inhomogeneous Lists	574
Implementing an Inhomogeneous List	576
Exercises	578
Solutions	580

Chapter 27 Multiple Inheritance 587

Multiply-Derived Classes	588
Multiple Indirect Base Classes	590
Virtual Base Classes	592
Constructor Calls	594
Initializing Virtual Base Classes	596
Exercises	598
Solutions	602

Chapter 28 Exception Handling 607

- Traditional Error Handling 608
- Exception Handling 610
- Exception Handlers 612
- Throwing and Catching Exceptions 614
- Nesting Exception Handling 616
- Defining Your Own Error Classes 618
- Standard Exception Classes 620
- Exercises 622
- Solutions 626

Chapter 29 More About Files 637

- Opening a File for Random Access 638
- Positioning for Random Access 640
- File State 644
- Exception Handling for Files 646
- Persistence of Polymorphic Objects 648
- Application: Index Files 652
- Implementing an Index File 654
- Exercises 656
- Solutions 660

Chapter 30 More About Pointers 681

- Pointer to Pointers 682
- Variable Number of Arguments 684
- Pointers to Functions 688
- Complex Declarations 690
- Defining Typenames 692
- Application: Dynamic Matrices 694
- Exercises 696
- Solutions 698

Chapter 31 Manipulating Bits 705

- Bitwise Operators 706
- Bitwise Shift Operators 708
- Bit Masks 710
- Using Bit Masks 712
- Bit-Fields 714
- Exercises 716
- Solutions 718

Chapter 32 Templates 721

- Function and Class Templates 722
- Defining Templates 724

Template Instantiation	726
Template Parameters	728
Template Arguments	730
Specialization	732
Default Arguments of Templates	734
Explicit Instantiation	736
Exercises	738
Solutions	742

Chapter 33 Containers 749

Container Types	750
Sequences	752
Iterators	754
Declaring Sequences	756
Inserting in Sequences	758
Accessing Objects	760
Length and Capacity	762
Deleting in Sequences	764
List Operations	766
Associative Containers	768
Sets and Multisets	770
Maps and Multimaps	772
Bitsets	774
Exercise	778
Solution	780

Appendix 783

Binary Numbers	784
Preprocessor Directives	787
Pre-Defined Standard Macros	792
Binding C Functions	793
Operators Overview	795
Operator Precedence Table	797
ASCII Code Table	798
Screen Control Sequences	800
Literature	801

Index 803

what() virtual method, 647
while statement
 structogram for, 96
 structogram for break within, 112
 syntax for, 97
Whitespace characters, 11
Width
 bit-fields, 715
width() method, 67, 491
Wordbyte union
 defining/using, 258
Write access
 open mode for, 638
write_at() method, 642

WriteError type exception, 651
write() method, 391, 392, 393
 of classes SepAcc and SavAcc, 648, 649
Write operation, 381
Writing
 blocks of records, 390
 characters, 75

X

XORoperator, 707

Z

Zero extension, 143